MILITARY SPECIFICATION

RESISTORS, VARIABLE, NONWIRE-WOUND

(ADJUSTMENT TYPE, LEAD-SCREW ACTUATED),

STYLE RJ24

THIS SPECIFICATION INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION (SEE 6.3). USE MIL-R-39035/2.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RJ24, adjustment type, lead-screw actuated, nonwire-wound, variable resistors. This style is available in characteristics C and F.

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids or request for proposal forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-22097 - Resistors, Variable, Nonwire-Wound (Adjustment Type), General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-22097, and as specified herein.

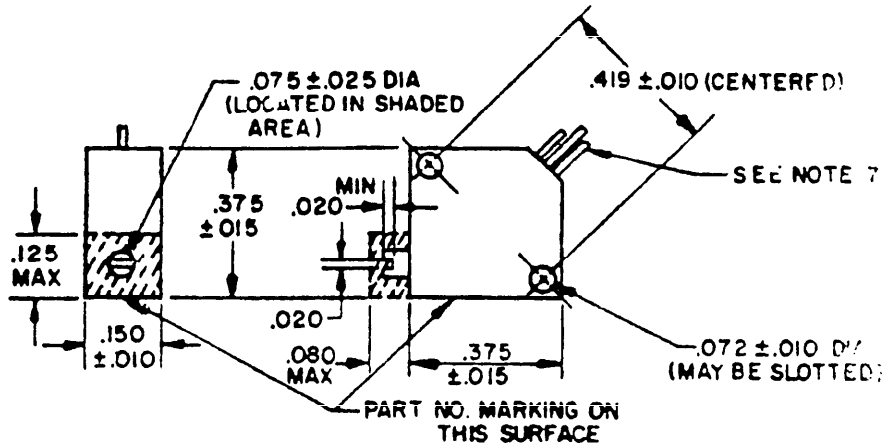
3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure.

3.3 Terminals. Characteristics C and F are available with P-type, W-type, X-type, and L-type terminals.

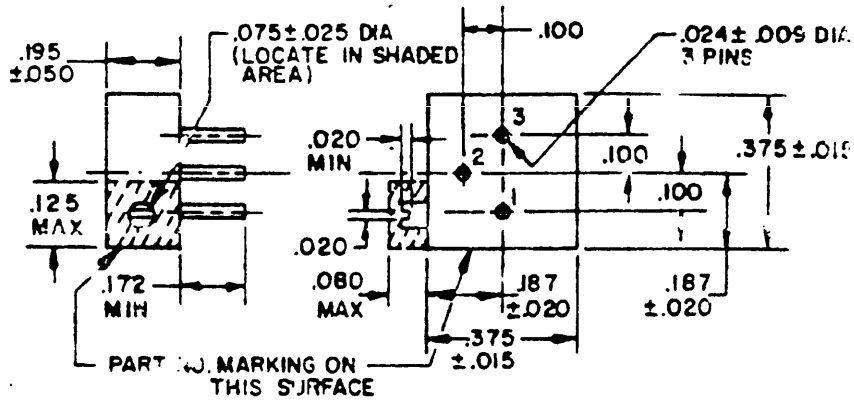
3.4 Power rating. The power ratings shall be 1/2 watt for characteristics C and F.

3.5 Nominal resistance value and maximum rated ac or dc working voltage. Nominal resistance values and maximum rated ac or dc working voltages shall be as specified in table I.

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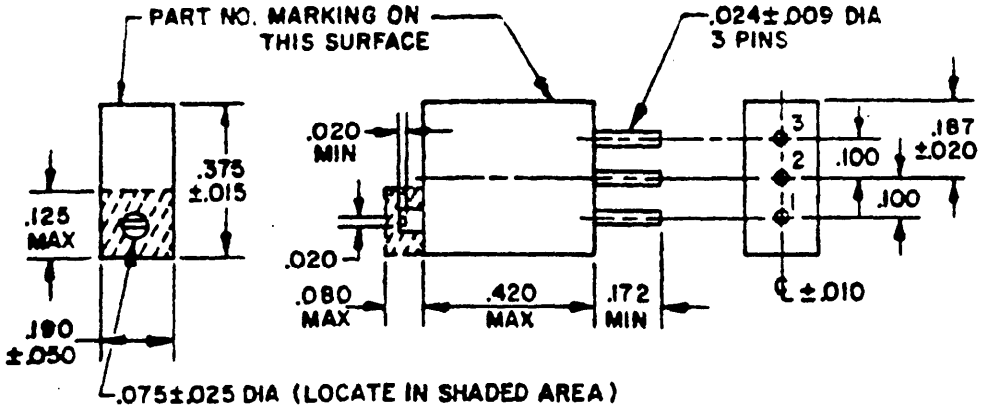
TERMINAL TYPE L



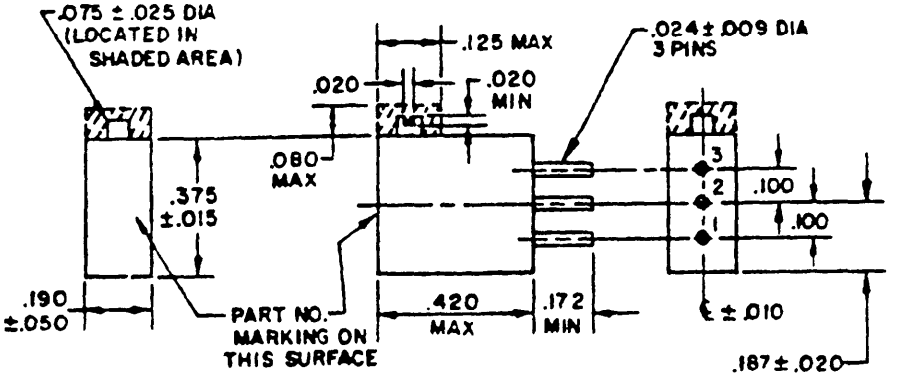
TERMINAL TYPE P

FIGURE 1. Style RJ24 resistors.

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TERMINAL TYPE W



INCHES	MM
.009	.23
.010	.25
.015	.38
.020	.51
.024	.61
.025	.64
.050	1.27
.072	1.83
.075	1.91
.080	2.03
.100	2.54
.125	3.18
.150	3.81
.172	4.37
.187	4.75
.190	4.83
.195	4.95
.375	9.53
.419	10.64
.420	10.67

TERMINAL TYPE X

NOTES:
1. The picturization of the styles above are given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.
2. The entire slot of the actuating screw must be above the surface of the unit.
3. For types P, W and X, normal mounting means is by use of pins only.
4. Unless otherwise specified, tolerance is ±.005(.13 mm).
5. Dimensions are in inches.
6. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
7. The three leads shall be stranded wire, AWG size 28 to 30, having a minimum length of 6(152.4 mm); they shall be insulated with polytetrafluoroethylene, stripped .250±.062(6.35±1.57 mm) from the end, and color coded.

FIGURE 1. Style RJ24 resistors - Continued.

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TABLE I. Nominal resistance value and maximum rated ac or dc working voltages.

Nominal resistance value	Maximum rated ac or dc working voltage per characteristic C, F
Ohms	Volts
10	2.23
20	3.1
50	5.0
100	7.0
200	10.0
500	15.8
1,000	22.3
2,000	31.6
5,000	50.0
10,000	70.7
20,000	100
25,000	111
50,000	158
Megohms	
0.10	223
0.25	300
0.50	300
1.0	300

3.6 Actual effective electrical travel. Actual effective electrical travel shall be 15 turns minimum, and 30 turns maximum.

3.7 Operating torque. Operating torque shall be a maximum of 5 ounce-inches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-22097 and as specified herein.

4.2 Dielectric withstanding voltage. The magnitude of test voltage shall be 900 volts at atmospheric pressure, and 250 volts at reduced barometric pressure.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-22097.

6. NOTES

6.1 The notes specified in MIL-R-22097 are applicable to this specification.

6.2 This specification supersedes MIL-R-22097/4B, 23 January 1970.

6.2.1 Characteristic B has been deleted. Resistors having characteristic B are replaced by resistors having characteristic C.

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6.3 Substitution data. The resistors specified herein are not for use in design after the date of this specification. They are authorized for use in design contracts effective prior to or on the date of this specification and to support existing military equipment. Resistors specified in MIL-R-39035/2 are preferred for design, and regardless of the failure rate designation can be used as substitutes for the inactivated resistors of the same resistance value, tolerance, and characteristic performance of this specification.

6.4 Weight. The maximum weight is 1.3 grams.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MI, MU, EL
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC, SH
Air Force - 19

Preparing activity:
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Agent:
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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